# ProtMistral-Tiny: Ultra-Compact Protein Language Modeling

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## Abstract

In this work, we introduce ProtMistral-Tiny, an ultra-compact autoregressive protein language variant of the Mistral architecture with just 15,000 parameters. Trained from scratch on a filtered subset of UniRef50 using residue-level tokenization, our model was benchmarked against two state-of-the-art baselines: ProtGPT2 (774M parameters) and ProGen2-Small (151M parameters). Despite extreme size constraints, ProtMistral-Tiny achieved structural scores comparable to ProGen2-Small. Notably, our model displayed an unusual pattern in sequence diversity: while local residue diversity was low, global diversity remained high, suggesting a conservative yet non-repetitive generation strategy. This behavior may arise from the model's limited capacity encouraging a more Markovian, stepwise generative process—eschewing memorized motifs. These results challenge existing assumptions about scale and capability in protein language modeling, demonstrating that even highly compressed models can produce structurally reasonable and diverse sequences.

## 1. Introduction

The advent of autoregressive language models has revolutionized natural language processing and recently expanded into the biological domain with remarkable success. These models, which predict tokens sequentially based on previous context, have evolved from early architectures like GPT (Ferruz et al., 2022) to more sophisticated implementations such as LLaMA (Lv et al., 2024). Their fundamental design—predicting the next element in a sequence given prior elements—has proven remarkably adaptable to protein sequence generation. Protein language models like ProtGPT2 (Ferruz et al., 2022) and ProGen2 (Nijkamp et al., 2023) leverage this autoregressive framework to generate novel protein sequences by treating amino acids as tokens in a specialized vocabulary. ProGen2, for instance, was trained on over 280 million protein sequences, enabling it to capture the complex distributional patterns of evolutionarily viable proteins. Unlike traditional protein engineering approaches that rely heavily on known structures, these models learn the implicit rules governing protein sequence space directly from data, offering a purely sequence-based approach to protein design. The success of these models stems from their ability to internalize the statistical patterns of natural proteins, effectively learning the grammar of protein sequences without explicit structural constraints.

The pursuit of de novo protein generation addresses crucial needs across multiple scientific frontiers. In therapeutic design, novel proteins offer potential solutions to targets previously considered "undruggable," with custom-designed binding proteins potentially overcoming limitations of traditional small molecules (Bennett et al., 2023). One of the primary advantages of LLMs has long been their advantage of scalability, easily being able to perform millions of virtual actions in a fraction of the time the same would take in the real world. This is particularly relevant in the biological domain, where processes like multiplex assays, pharmaceutical trials, and large-scale chemical screening have traditionally been bottlenecked by the immense resources required to conduct exhaustive explorations. Critically, one discovery from the field of protein design is that the lossy nature of these operations can be outpaced by their sheer scale (Bennett et al., 2023). Despite this, current autoregressive pLMs operate in a high-compute regime, motivated by similar scaling laws to those governing natural language processing. The largest such models, such as Evo, despite being especially successful at understanding natural sequences, can reach up to 7 billion parameters, arguably negating the advantages that make virtual assays so attractive in the first place (Nguyen et al., 2024).

While large models are undeniably powerful, for purposes such as search or other NP-hard explorations of chemical space, we reason that twenty good-enough guesses may in

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fact be superior to one fantastic one. We motivate our paper as an attempt to build a model optimized for such a task. While this paper does not aim to buck scaling laws, we proceed in the opposite direction, instead aiming at producing a minimalist autoregressive pLM optimized for inference speed and data efficiency. To do so, we aim to leverage advances in efficient architectures in natural language processing, adapting a tiny variant of the Mistral 7B architecture (MistralAI, 2024) for both its linear attention complexity and efficient inference. In contrast to models such as ProGen2 and Evo which focus primarily on scaling to massive amounts of biological data (by nature accepting a drop in annotation quality) and parameters, we focus more on leveraging architectural improvements from other domains to boost raw compute-per-token efficiency.

## 2. Methods

### 2.1. Dataset Curation

We trained on a subset UniRef50 database for our training process (The UniProt Consortium, 2024). UniRef50 is built by clustering UniRef90 seed sequences that have at least 50% sequence identity to and 80% overlap with the longest sequence in the cluster. As UniRef90 was already a reduced version of UniRef100 based on clustering, we did not cluster, rather using the random split provided on Hugging Face. We took advantage of the Mistral architecture's sliding window attention, which has an O(n) token complexity, as well as our small model size (allowing a higher proportion of VRAM for data), to train on very large proteins with only one A100 GPU. As such we filtered out sequences larger than 10k amino acids, which removed 2790 sequences out of 59 M training sequences and 1 and 0 sequences respectively from the 5k testing and validation set. We further subsampled a random 100k of the training set and a static 500 validation and test sequences each to maintain the original data balance. For comparison, this is roughly 450,000% smaller than ProtGPT2's train set. Unlike ProtGPT2 or ProGen2, we used residue level tokenization, adopting ESM-2's tokenization scheme. This includes the 20 amino acids, the standard beginning/end of sequence/pad tokens, and a token to map unknown characters. These datasets were processed (and thus tokenized) with variable context lengths.

### 2.2. Model Architecture

We adopted the tiny-Mistral architecture, a tiny variant of the Mistral 7B architecture (Jiang et al., 2023). Our model is a 2-layer decoder with 15k parameters, 10,067% smaller than ProGen2-Small and 51,600% smaller than ProtGPT-2. The Mistral architecture is well suited to tiny variants due to it's adoption of grouped-query attention (GQA), and sliding window attention (SWA). GQA groups heads together, having e.g. each of 2 K/V head queried by 2 Q heads (in our tiny-Mistral specifically), effectively 'recycling' more heads for half the cost. This design choice is paired with focusing more parameters on the feed-forward intermediates than attention heads, leading to a model which is more data-efficient per FLOP, effectively acting as a deeper model. Despite this, we only partially took advantage of this, finding a latent size of 32 to be surprisingly effective.

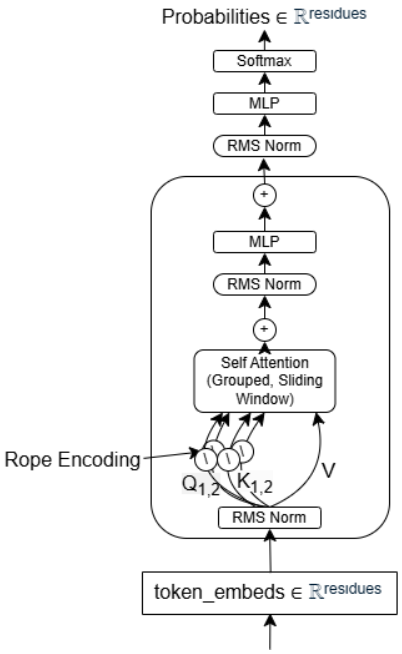



*Figure 1.* Model Architecture

We used Mistral's linear attention complexity - produced by using a sliding attention window over the tokens (SWA) - to train on a more diverse set of protein sizes even when memory usage is very limited (e.g. small-compute regimes). That said, we adhered to a fairly large window size (4096) to avoid constraining performance. We maintained all hyperparameters of tiny-Mistral except for tokenization - despite this, our model was 13,300% smaller than the NLP variant, owing to the smaller vocabulary and thus encoder layer.

### 2.3. Training

We trained for 10 epochs with a batch size of 2, a relatively standard learning rate of 2e-4, an AdamW optimizer, and standard cross-entropy loss. We incorporated early stopping with patience of 3 and an improvement threshold of 0.001, which began to trigger by the last epoch - relatively unsurprising given we are likely in a higher-parameter regime. We come to this conclusion from the fact that we decrease data size more than parameter count proportionally - thus large pLMs train on a lower parameter-to-token ratio (see prior discussion of ProtGPT-2 data & parameter size relative to our model). We experimented with cosine-scheduling but

found it slightly inferior to no schedule at all. Batch size was maintained in part due to compute constraints and in part due to our training on especially large sequences (which scale poorly with higher batch sizes) but could very likely be increased for other applications.

These choices allowed us to reduce our train time to a single A100 GPU for 3 hours - an especially small compute budget relative to competing models (ex: 2048% fewer A100 hours than ProtGPT-2). This was motivated by our goal of producing an aggressively lightweight model, both at inference and training.

### 2.4. Metrics

In structure-based metrics, the predicted local distance difference test (pLDDT) measures confidence in the accuracy of each residue's position by comparing atom-pair distances in the predictions with those from the reference. For residue $i$, pLDDT is defined as:

$$\text{pLDDT}_i = 100 \left(1 - \frac{|d_{\text{pred}}(i) - d_{\text{ref}}(i)|}{d_{\text{ref}}(i)}\right)$$

where $d_{\text{pred}}(i)$ and $d_{\text{ref}}(i)$ are the distances between atom pairs in the predicted and reference structures, respectively. The mean pLDDT score is taken across all residues in a sequence.

The TM score is a measure of the global structural alignment between two structures, specifically how similar the overall fold of a predicted structure is to a reference structure. This score is less sensitive to local inaccuracies (like small misplacements of side chains) and focuses more on the overall fold. The TM-score is defined as:

$$\text{TM-score} = \max \left[\frac{1}{L_{\text{ref}}} \sum_{i=1}^{L_{\text{align}}} \frac{1}{1 + \left(\frac{d_i}{d_0}\right)^2}\right]$$

where $d_i$ is the distance between the $i$-th pair of residues in the aligned structures, $L_{\text{align}}$ is the number of aligned residues, $L_{\text{ref}}$ is the length of the native reference structure, and $d_0(L_{\text{ref}})$ is a scale normalization factor typically defined as: $d_0(L_{\text{ref}}) = 1.24(L_{\text{ref}} - 15)^{1/3} - 1.8$ but often simplified to a constant such as 5Å (Zhang & Skolnick, 2004).

However, because a true reference structure was not available for our model, we used a predicted TM-score (pTM), which was generated by a neural network head trained to estimate the TM-score using internal model features such as the pAE matrix. This learned pTM-score represents the global structural accuracy of the predicted fold, providing a structural confidence estimate even in the absence of an experimental baseline.

The predicted aligned error (pAE) estimates the expected error in residue-to-residue distances, calculated as:

$$\text{pAE} = \frac{1}{N} \sum_{i=1}^{N} |d_{\text{pred}}(i) - d_{\text{ref}}(i)|$$

where $N$ is the number of residues in the structure. Lower pAE values indicate better predicted accuracy.

Together, pLDDT, pTM, and pAE offer complementary views: pLDDT quantifies per-residue confidence, pTM evaluates overall fold reliability, and pAE reflects the predicted residue-to-residue positioning error.

For sequence diversity, we employed three complementary metrics: Shannon entropy, edit distance, and Jaccard similarity. Shannon entropy at residue position $i$ is defined as:

$$H_i = -\sum_{a \in A} p(a_i) \log p(a_i)$$

where $A$ is the set of possible amino acids, and $p(a_i)$ is the probability of amino acid $a$ at position $i$. Higher entropy indicates greater variability. Edit distance between two sequences $S_1$ and $S_2$ is:

$$\text{EditDist}(S_1, S_2) = \min(\text{insertions}, \text{deletions}, \text{substitutions})$$

This measures how many operations are required to transform one sequence into another. Jaccard similarity between sequences $S_1$ and $S_2$ based on k-mers of length $k$ is:

$$\text{JaccardSim}(S_1, S_2) = \frac{|\text{K-mers}(S_1) \cap \text{K-mers}(S_2)|}{|\text{K-mers}(S_1) \cup \text{K-mers}(S_2)|}$$

where K-mers are subsequences of length $k$, and a higher Jaccard similarity means more shared local motifs, implying less diversity (with the caveat that in proteins motifs may be reasonable or even desired).

Collectively, these metrics capture sequence variability: entropy assesses diversity per position, edit distance quantifies global differences, and Jaccard similarity measures local motif overlap. Evaluating all three provides a comprehensive view of sequence diversity across generated proteins.

## 3. Results

Structurally, ProGen2-Small and ours appeared to be quite similar, with scores around 47 indicating a moderate uncertainty about residue placement. By contrast, ProtGPT2 was

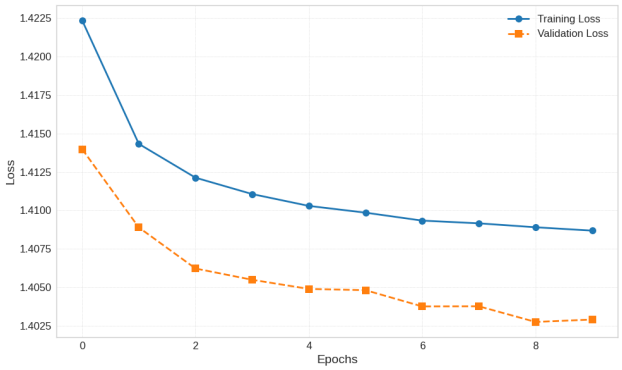



*Figure 2.* Training ProtMistral-Tiny on a 100k Subset of UniRef50

| Model | pLDDT | pTM | pAE |
|---|---|---|---|
| ProtGPT2 | 66.07 | 0.35 | 14.22 |
| ProGen2-Small | 47.65 | 0.29 | 17.68 |
| ProtMistral-Tiny (Ours) | 47.15 | 0.28 | 16.83 |

*Table 1.* Structure metrics for different autoregressive pLMs, computed via AlphaFold; sample of 200 proteins, max-length 100.

more confident, achieving a pLDDT of 66.07. A similar trend occurred in pTM and pAE, with comparable values for ProGen2 and our model, with ProtGPT2 leading on both local and global structure prediction accuracy. This is most obviously a function of parameter count – though it does also suggest that our model is comparable to ProGen2, a model thousands of times larger. Alternatively, ProGen2's poor performance could be due to incorporation of rarer data sources (e.g. metagenomics, less conserved/stable proteins), with ours roughly adhering to scaling laws, though this seems less likely.

| Model | Entropy | Edit Dist. | Similiarity |
|---|---|---|---|
| ProtGPT2 | 15.17 | 51.41 | 0.0081 |
| ProGen2-Small | 26.50 | 84.48 | 0.0109 |
| ProtMistral-Tiny (Ours) | 8.05 | 78.96 | 0.0077 |

*Table 2.* Sequence diversity metrics for different autoregressive pLMs; sample of 200 proteins, max-length 100.

ProGen2-Small appeared to generate highly novel sequences with an alphabet of common motifs, as indicated by its high Shannon Entropy, Edit Distance, and Jaccard Similarity (or k-mer repetition). This is undercut by its significantly worse structural metrics, indicating that this novelty often forced a tradeoff of a quality or stability. By contrast, the ProtGPT2 sequences were more stable while being only moderately surprising, not relying strongly on k-mers (lower Jaccard Similiarity). By contrast, our model has something of an odd result, being locally very conservative, not relying on k-mers, yet simultaneously being globally diverse.

We hypothesize that this may be due to the fact that, even with similiarity clustering, many sequences are repetitive, whether from transposons, tandem repeats, or conserved functional motifs. The ways in which nature does often repeat itself may be better learned by these larger models. Since our model does not learn these motifs as closely, it seems to predict less heterogenous proteins than larger models (ex: less motif-blocks, repetitive domains) - this may explain why its results are locally homogenous, as demonstrated by our low Shannon Entropy and Jaccard Similiarity values. Defaulting to these motifs/patterns may constrain model predictions, whereas a smaller model is in some sense forced to lay down the train tracks in front of itself without planning too far ahead, in a Markov-chain-like fashion. Promisingly, this does not seem to significantly decrease pLDDT compared to ProGen2, which does nearly the opposite, suggesting that these train-track proteins lie near to the viable manifold despite this behavior.

The standout result of our analysis was the shocking competence of our model relative to its larger counterparts, as seen in scores in the similar structural scores to the thousands of times larger ProGen2-Small. Notably, ProtMistral-Tiny also possessed significant sequence diversity, indicating the structural "reasonability" of these sequences (as predicted by AlphaFold pLDDT) was not solely due to memorization. These proteins are smaller than our sliding window (4k), so a small-sequence inductive bonus from locality also cannot be responsible - neither can residue tokenization, which was shared with ProtGen2. Our performance seems partly attributable to the increased data-efficency of the Mistral architecture, arising e.g. GQA which allows recycles a deeper model out of less parameters. This suggests the baseline competence encoded by these models is significantly more compressible than previously thought.

## 4. Conclusion

Our work explores how far a lightweight autoregressive model can go in the realm of protein sequence generation. Despite being several orders of magnitude smaller than ProtGPT2 and ProGen2-Small, our 15K-parameter Mistral-based model was able to generate protein-like sequences that scored competitively to ProGen2-Small on structural metrics. While we underperformed both larger models in absolute terms, in parameter efficiency this is a deviation from what scaling laws would typically predict, beyond what we would naively predict our model to be capable of. For instance, GQA seems like it should at most "double" model size, yet we perform as though having a parameter count significantly beyond that. We theorize that this may be due to the repetitive nature of sequence data, or a Markov-like quirk of small language models that is suprisingly effective on (at least small) sequence data - either way, it remains a

promising direction for future research.

The primary constraints with our approach were the compute constraints of AlphaFold metric comparisons, as well as the absence of controls to compare to determine the cause of results. Moving to ESMFold and running a much broader sweep of tests would be a logical future work, as would ablating various components of the model to see which components contribute most heavily to its apparent efficacy. Similarly, more diverse metrics (e.g. real world validation against a well-established reference) are outside of the scope of this project but would make an interesting test for the 'Markov-like efficacy' hypothesis. Since this approach explicitly is aiming for parameter efficiency, we do not consider limited compute or model size to be significant constraints.

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